

BOURNEMOUTH INTL, United Kingdom

WMO: 038620

Lat: 50.7794N

Lon: 1.8361W

Elev: 38

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.3	26.6	20.6	15.5	27.6	23.1	17.4	30.2	30.2	49.7	27.4	49.6	3.4	350	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	78.9	65.7	75.4	63.9	72.6	62.7	67.2	75.8	65.5	72.7	64.0	70.1	10.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.2	90.0	69.7	62.8	85.6	67.9	61.5	81.9	66.6	31.6	75.9	30.3	72.9	29.2	70.3	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.6	21.8	19.5	18.4	84.1	2.8	3.9	16.4	86.9	14.8	89.2	13.2	91.3	11.1	94.1		
			17.8	70.4	2.8	2.3	15.8	72.0	14.2	73.4	12.6	74.7	10.5	76.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	41.5	41.7	44.4	48.1	53.6	58.8	62.6	62.1	57.7	52.7	46.4	42.4
	DBStd	9.27	6.23	5.77	5.05	5.13	4.93	4.23	4.03	4.08	4.20	5.45	6.07	6.84
	HDD50	1221	268	235	183	92	20	0	0	0	2	36	138	247
	HDD65	5136	729	653	640	506	354	191	97	108	220	381	557	700
	CDD50	1612	4	3	8	36	132	265	391	376	234	120	31	12
	CDD65	51	0	0	0	0	1	6	23	19	2	0	0	0
	CDH74	453	0	0	0	1	13	66	225	137	11	0	0	0
	CDH80	68	0	0	0	0	0	7	36	25	0	0	0	0
Wind	WSAvg	9.1	10.4	10.1	9.7	9.1	9.1	8.7	8.3	8.0	8.0	8.9	9.1	9.7
Precipitation	PrecAvg	32.0	3.6	2.5	2.4	2.1	2.1	2.2	1.8	2.2	2.7	3.4	3.7	3.6
	PrecMax	55.0	9.0	7.2	5.3	5.9	5.3	8.7	5.4	5.4	8.8	8.5	9.3	8.2
	PrecMin	20.0	0.4	0.2	0.0	0.0	0.4	0.0	0.2	0.2	0.3	0.1	0.7	0.9
	PrecStd	5.9	1.9	1.7	1.2	1.4	1.3	1.5	1.1	1.3	1.8	2.0	2.0	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.9	55.7	61.9	72.0	75.9	80.6	84.3	83.4	75.4	67.7	60.8	57.1
		MCWB	52.7	51.1	52.2	58.8	62.6	66.0	66.7	68.7	65.1	61.4	58.0	54.7
	2%	DB	53.8	53.3	57.8	64.3	71.2	75.1	79.7	77.6	71.6	64.6	58.7	55.6
		MCWB	51.9	50.4	51.2	54.5	60.5	63.2	65.7	65.8	62.9	59.4	56.4	53.7
	5%	DB	52.3	51.8	54.8	60.3	67.4	71.6	76.0	73.8	68.6	62.8	57.4	54.5
		MCWB	50.4	49.2	49.6	52.7	58.8	61.6	64.3	63.9	60.8	58.9	55.0	52.6
	10%	DB	51.0	50.5	52.8	57.3	63.5	68.6	72.7	71.0	66.4	61.3	56.0	53.1
		MCWB	49.0	48.1	48.8	50.7	56.2	60.0	62.8	62.4	59.9	57.9	53.6	51.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.7	52.9	54.1	59.6	65.2	68.0	69.2	70.5	66.5	62.6	58.8	55.5
		MCDB	54.4	53.9	58.2	68.9	72.5	77.6	79.7	79.6	72.2	65.5	59.7	56.3
	2%	WB	52.2	51.3	52.2	55.9	61.9	64.9	66.9	67.4	64.3	61.2	56.9	54.3
		MCDB	53.3	52.6	55.4	63.0	68.9	72.6	76.9	74.9	69.2	63.4	58.2	55.4
	5%	WB	50.8	50.0	51.0	53.7	59.4	62.7	65.1	65.4	62.8	59.9	55.5	53.0
		MCDB	52.1	51.2	53.7	58.4	65.7	69.7	73.5	71.5	66.5	62.0	57.1	54.3
	10%	WB	49.3	48.5	49.8	52.0	57.2	60.9	63.6	63.8	61.5	58.7	53.9	51.3
		MCDB	50.8	50.1	52.2	55.9	62.3	66.8	70.7	68.8	65.0	60.8	55.8	52.8
Mean Daily Temperature Range	5% DB	MDBR	11.5	12.4	14.8	16.7	16.9	16.8	17.2	17.1	17.3	14.5	12.9	11.5
		MCDBR	10.5	11.9	17.7	21.9	23.2	22.4	24.2	21.8	20.1	14.2	11.1	9.0
		MCWBR	10.1	9.8	12.4	13.7	13.9	12.6	12.4	11.6	12.8	10.3	9.6	9.1
	5% WB	MCDBR	10.4	10.7	14.7	18.5	20.9	19.5	20.8	18.4	17.1	12.2	10.2	9.2
		MCWBR	10.4	9.8	11.1	12.3	13.4	11.8	11.9	11.0	12.0	10.1	9.8	9.6
Clear-Sky Solar Irradiance	taub	0.332	0.347	0.383	0.400	0.403	0.398	0.391	0.384	0.373	0.361	0.333	0.322	
	taud	2.366	2.377	2.276	2.256	2.289	2.333	2.357	2.397	2.416	2.425	2.417	2.350	
	Ebn at Noon	217	247	257	266	271	273	272	269	259	239	216	200	
	Edh at Noon	21	26	34	39	40	38	37	34	30	25	19	18	
All-Sky Solar Radiation	RadAvg	261	490	873	1380	1666	1832	1737	1453	1082	610	321	205	
	RadStd	21	57	111	130	95	175	136	112	77	49	34	24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (69 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page